Learning how to prove Workshop Week 9

Groups

Group Homomorphisms

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1. Let $G_1 \times G_2$ be the direct product of two groups G_1 and G_2. We define

$$\begin{array}{lll} \pi_1 : G_1 \times G_2 \to G_1 & \text{by} & \pi_1(g_1,\, g_2) = g_1 \\ \iota_1 : G_1 \to G_1 \times G_2 & \text{by} & \iota(g_1) = (g_1,\, 1_{G_2}) \end{array}$$

   Prove

   (a) π_1 is an epimorphism, called the **projection onto** G_1.

   (b) ι_1 is a monomorphism, called the **injection of** G_1 **onto** $G_1 \times G_2$**.**

   Similarly, we can define a **projection onto** G_2, and an **injection of** G_2 **onto** $G_1 \times G_2$.

2. Let n be a nonzero integer, and consider the subgroup $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ of $\mathbb{Z}$.

   (a) Prove that $\mathbb{Z} \cong n\mathbb{Z}$.

   (b) Show that $n\mathbb{Z} \cong m\mathbb{Z}$ for all $n \neq 0$ and $m \neq 0$.